Engineering Mathematics - I · Unit_01 · Unit 1

DRAFT — UE26MA141A · Unit_01 · Unit 1 · Problems on Composite and Implicit functions

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Problems on Composite and Implicit functions

1 Problems on Composite and Implicit functions

Study focus: Problems on Composite and Implicit functions. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000110 · EV0000111

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We will solve problems on composite functions.

Key details

- Then find the value of do u by dou t.
- It means u becomes the composite function of r and t.
- dou u by dou t equals dou u by dou x into dou x by dou t plus dou u by dou y into dou y by dou t.
- Then dou u by dou t becomes 2x into cos of x square plus y square into minus r sin t plus 2y cos of x square plus y square into r cos t.
- Now because u is a composite function of r and t, we eliminate.
- Let is x and y.
- R sin t into cos c.
- Absolutely problem here.
- U is a function of x and y.
- X and Y are the functions of R and T.
- Find dou u by dou theta where u equals x square y plus y z q, x equal to r cos theta, y equal to r sin theta, and z d equals to theta.
- In this problem, u is a function of x y z.
- where xy z are the functions of r and theta it means u becomes the composite function of r and theta so in this problem we want only one particular case that is dou u by dou theta so when we use chain rule.
- W by dou y is x square plus z power three.
- W u by dou z equals three y z square.

Example or application from the lecture

- Let us look at the first problem.
- It is given that u equals sin of x square plus y square where x equal to r cos t and y equal to r sin t.
- It is given that u is a function of x and y, where x and y are the functions of r and t.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000110 · EV0000111 · EV0000112 · +1 source units in pack.json

Core explanation · 05:00–09:59

[ASR low confidence; check audio].

Key details

- If u equal to a function of r where r equal to square root of x square plus y square plus z square, then express dou u by dou x in terms of f dash of r.
- When r is a function of x, y, and z.
- Because r is square root of x square plus y square plus z square, so the partial derivative of r with respect to x is going to be x divided by square root of x square plus y square plus square is after calculation up to 12.
- And x and y are the functions of r and theta, then u becomes the composite function of r and theta.
- So what would be the next step in this particular problem?
- Can you stop here?
- Because u is a composite function of r and theta, we need to eliminate this x and y.
- Zero because after cancellation of this and this.
- If u equal to tan in us y by x, when x and y are the functions of r and theta, then show that 2 u by 2 theta equals 1.
- Again after writing the particular chain.
- Then when we compute dou u by do x, dou u by dou y, then dou x by dou theta and dou y by dou theta and finally upon using x and y to be r cos and r sign respectively.
- We see that two by dou theta becomes one.
- U is a function of x and y.
- When x and y are the functions of s and t then u becomes the composite function of s and t.

Example or application from the lecture

- Let us look at this particular problem.
- Again, when we look at the chain, it is given that u is a function of r.
- So when we write the chain rule for dou u by dou x, it be it becomes dou u by dou x equals dou by dou r into dou r by dou x so because u is f of r so dou u by dou r becomes f dash of r into dou r by dou x so let us find the value of dou r by dou x.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000114 · EV0000115 · EV0000116 · +1 source units in pack.json

Core explanation · 10:00–14:59

And you must know why did we use x equal to e power s and y equal to s plus t after this particular step.

Key details

- So because u is a composite function of s and t.
- Then find the value of dou by dou x and dou by dou y.
- U is a function of r and theta where r and theta functions of x and y.
- Then u becomes composite function of x and y.
- Then when we use chain rule, we get dou u by dou x has this and dou u by dou y has this so after solving for dou r by dou x dou dita by dou x dou y and dou dita by dou y we get this and this as the answer for this particular problem.

Example or application from the lecture

- It is given that u equals f of r theta, where r equals square root of x square plus y square and theta equals that inverse y by x.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000118 · EV0000119 · EV0000120 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Problems on Composite and Implicit functions transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/004_Problems_on_Composite_and_Implicit_functions__d1ea652fce3206d2.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Problems on Composite and Implicit functions transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/004_Problems_on_Composite_and_Implicit_functions__d1ea652fce3206d2.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Problems on Composite and Implicit functions transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/004_Problems_on_Composite_and_Implicit_functions__d1ea652fce3206d2.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9982; only the JSON segmentation supplies the displayed timestamps.
Problems on Composite and Implicit functions transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/004_Problems_on_Composite_and_Implicit_functions__d1ea652fce3206d2.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
2. 2 video-to-slide associations were not promoted to a match. Candidate rankings are listed in the transcript corpus and video-slide-matches.csv; the slide decks appear separately.
3. 1 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
4. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
5. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
6. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
7. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00026	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0017. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0018 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0042. Sidecar variants compare...
SRC00027	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0017. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0018 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0042. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00028	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0017. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0018 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0042. Sidecar variants compare...
SRC00029	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0017. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0018 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0042. Sidecar variants compare...

C Revision and retrieval practice